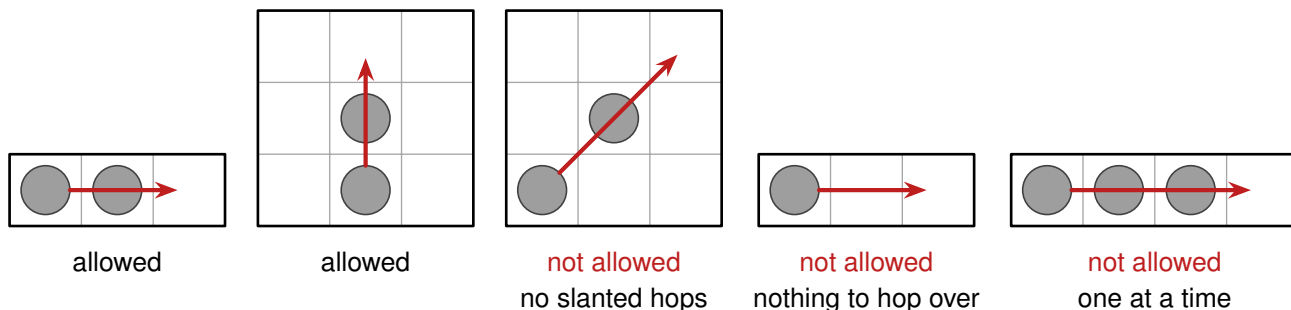


# The Hop, and a Strip

The pennies are soldiers. A soldier may **hop over a neighbor** (up, down, left or right) into the **empty** square directly beyond; the soldier that was hopped over **goes home** (take it off). That is the only move. Hops may go in any direction, including backwards.



## Warm-up: leave one

Put a penny on every gray circle and hop until one soldier is left. **One of the three is impossible** — which one? Can you *prove* it, not just fail at it?

**S1**

hops: \_\_\_\_\_

**S2**

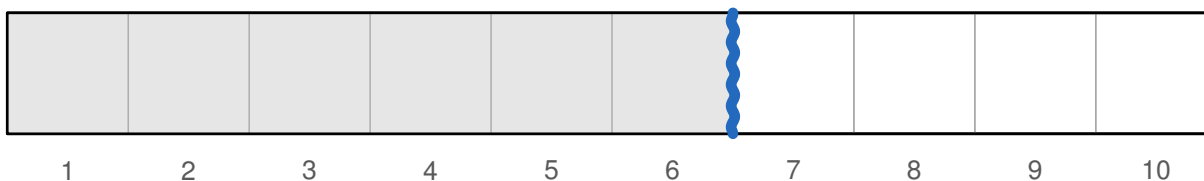
hops: \_\_\_\_\_

**S3**

hops: \_\_\_\_\_

## How far along a strip?

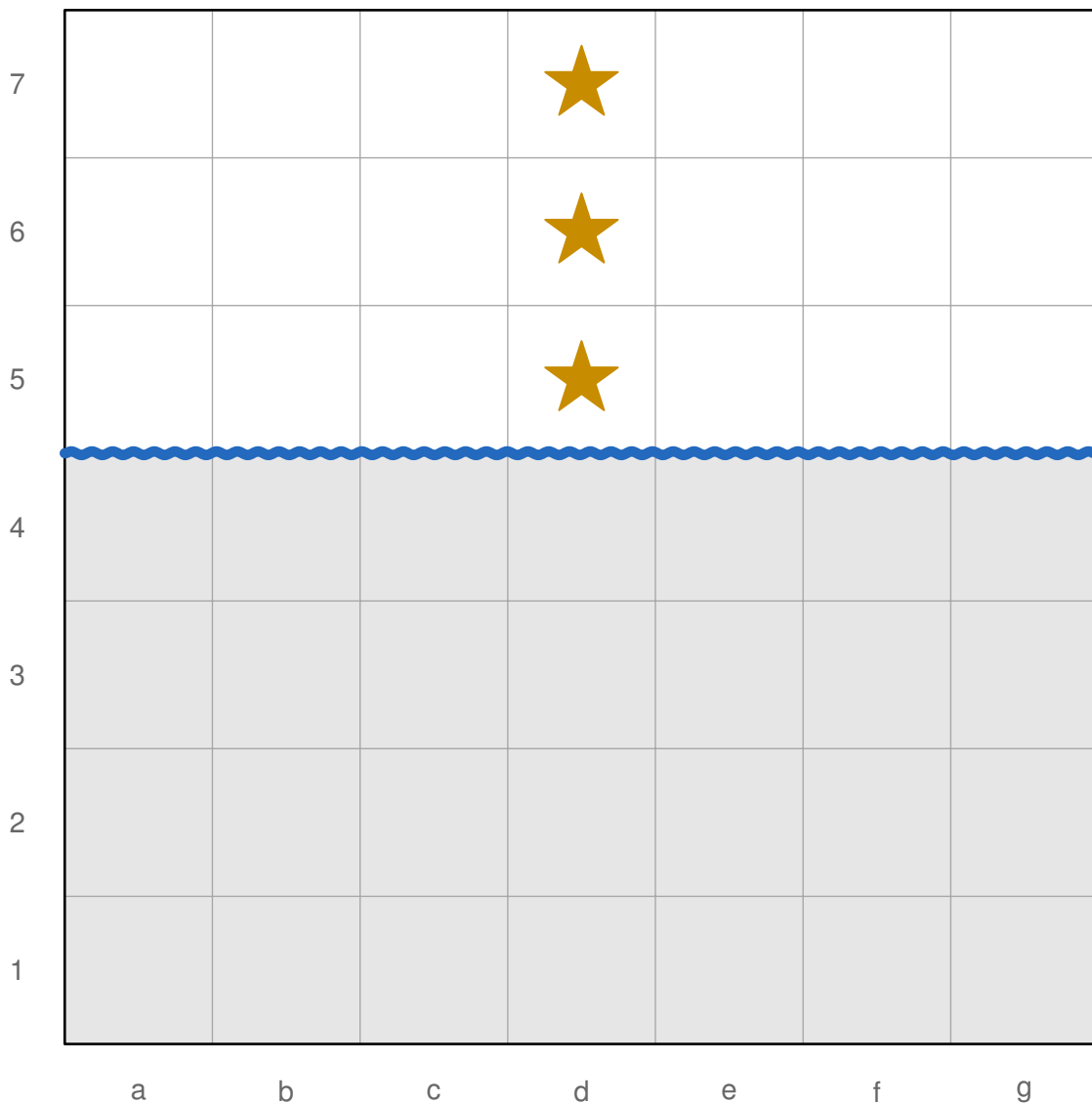
The gray squares 1–6 are the camp; the river runs between squares 6 and 7. Your army may stand on *any* gray squares you like — as many soldiers as you want, gaps allowed.



- Get a soldier to square 7. (Easy. Which two squares must the soldiers start on?)
- Get a soldier to square 8. Try many armies: solid blocks, blocks with gaps, big armies.
- After some tries: do you believe square 8 is *impossible*? A proof is waiting on page 3 — but first, page 2.

# Across the River

Same game, two dimensions: the camp is gray, the river is blue, every square north of it is empty. **How far north can one soldier get, and how few soldiers does it take?** Reach the lowest star (one row north), then the middle star (two rows), then the top star (three rows). Each time find the **smallest army** that works, and record it (a hop is *from*→*to*, like d3→d5).



**1 row:** fewest \_\_\_\_\_ army \_\_\_\_\_ hops \_\_\_\_\_→\_\_\_\_\_

**2 rows:** fewest \_\_\_\_\_ army \_\_\_\_\_ hops \_\_\_\_\_→\_\_\_\_\_→\_\_\_\_\_→\_\_\_\_\_

**3 rows:** fewest \_\_\_\_\_ army \_\_\_\_\_ hops \_\_\_\_\_→\_\_\_\_\_→\_\_\_\_\_→\_\_\_\_\_

\_\_\_\_\_→\_\_\_\_\_→\_\_\_\_\_→\_\_\_\_\_→\_\_\_\_\_

## Prove it

*Three soldiers can never get two rows north. Prove it. (Work backwards: the soldier that lands two rows north hopped from where, over whom? So what was true just before? And how did *that* come about?)*

# Prices: Why Square 8 Is Impossible

Back to the strip. We will give every square a **price**, and watch what happens to the total price of the army when a soldier hops. Square 8 — the square we are trying to reach — costs \$1. Going left, each square costs less:

|                 |      |      |      |      |      |      |      |            |      |      |
|-----------------|------|------|------|------|------|------|------|------------|------|------|
| ...             | .034 | .056 | .090 | .146 | .236 | .382 | .618 | <b>\$1</b> | .618 | .382 |
| .021, .013, ... | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8          | 9    | 10   |

- The price rule.** Each price is the *sum of the two prices to its left*:  $.382 = .236 + .146$ . Check three more. (Prices are rounded to three decimals, so one of them is off by .001.) To the right of square 8 the prices go back down by the same rule, mirrored; to the left they go on forever: .021, .013, .008, .005, ...
- A hop never makes the army richer.** A soldier on square  $k$  hops over square  $k+1$  onto square  $k+2$  (toward the target). Before the hop the two soldiers were worth  $p_k + p_{k+1}$ . After it, the one soldier is worth  $p_{k+2}$ . Compare, using the price rule. Now do a hop *away* from the target: a soldier on  $k+2$  hops over  $k+1$  onto  $k$ . Before:  $p_{k+2} + p_{k+1}$ . After:  $p_k$ . Which is bigger? (Use  $p_{k+2} = p_{k+1} + p_k$ .)
- The whole camp is worth exactly \$1.** Add up the camp, starting at square 6 and walking left. Keep a running total:

|               |      |      |      |      |      |      |      |      |      |
|---------------|------|------|------|------|------|------|------|------|------|
| square        | 6    | 5    | 4    | 3    | 2    | 1    | 0    | -1   | -2   |
| price         | .382 | .236 | .146 | .090 | .056 | .034 | .021 | .013 | .008 |
| running total | .382 | .618 |      |      |      |      |      |      |      |

What number is the running total creeping toward? Does it ever get there? (Look at how far the total is from \$1 after each step, and compare with the prices.)

- The punchline.** Your army is finite, so it uses only some of the camp squares: it is worth *less* than \$1. Hops never raise the total. But a single soldier standing on square 8 is worth \$1 all by himself. So... finish the sentence.
- Where does .618 come from?** It is the number  $x$  with  $x + x^2 = 1$ . Check:  $.618 \times .618 \approx .382$ , and  $.618 + .382 = 1$ . (This number is famous. Ask your table leader its name.)

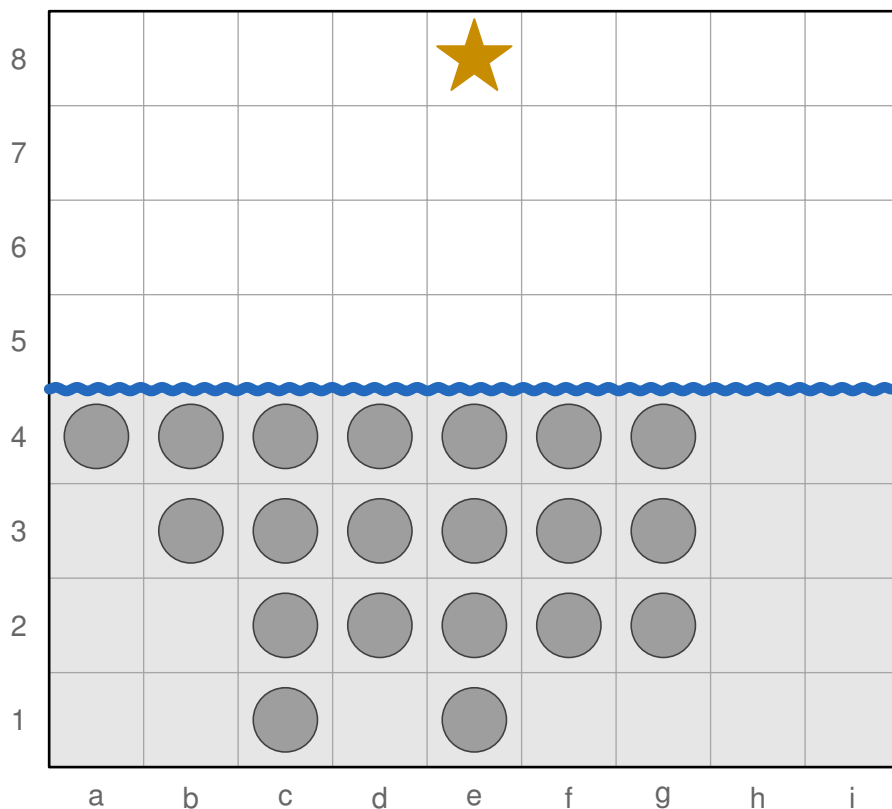
## Two dimensions

Give every square on the big board a price too: a target square gets \$1, and each other square gets the price of the square that many *walking steps* away on the strip. With the target **five rows north** of the river, the entire endless camp adds up to exactly \$1 again — the calculation is in your table leader's notes. Conclusion: **no army, however large, can ever send a soldier five rows north**. Four rows is the limit.

# Four Rows North — and Never Five

## The 20-soldier army

Four rows north is possible, and the smallest army that can do it has 20 soldiers (a 2007 computer search showed 19 is not enough). Here is one such army. Set it up on a big grid sheet — the target is column e, four rows past the river — and find the 19 hops. Hint: the last hop is e6→e8, so at the end you need soldiers on e6 and e7; a soldier arriving on e7 comes from e5, and so on.



Hops: \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_  
 \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_  
 \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_  
 \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_ → \_\_\_\_\_

## The whole story

|                         |   |   |   |    |                     |
|-------------------------|---|---|---|----|---------------------|
| rows north of the river | 1 | 2 | 3 | 4  | 5                   |
| fewest soldiers         | 2 | 4 | 8 | 20 | impossible, forever |

On a strip the limit is **one** square past the river; on the plane it is **four** rows. Both limits are proved by the same trick: prices that no hop can increase, adding up to exactly \$1 over the whole endless camp. The game is called *Conway's Soldiers*, after the mathematician John Horton Conway, who found the proof in 1961.